COMP-SCI 5520: Network Architecture I
Syllabus, Fall 2026

Instructor: Dr. Maya Okafor. Office hours: Tuesdays and Thursdays 2:00-3:30 pm, or by appointment through RooRoom. Lectures: Mondays and Wednesdays 4:00-5:15 pm. Questions about assignments should go through the RooRoom question pool so that one answer reaches the whole class.

Course description

This course covers the architecture of computer networks: the layered model, the application layer (HTTP, DNS), the transport layer (UDP, TCP reliability, flow control and congestion control), the network layer (IP addressing, routing algorithms, OSPF and BGP), and the link layer. A semester-long networks project has students build and measure a reliable transport protocol in three parts.

Grading

- Homework assignments (5): 25%
- Quizzes (4): 10%
- Networks project, Parts 1-3: 30% (Part 1 5%, Part 2 10%, Part 3 15%)
- Midterm exam: 15%
- Final exam: 20%

Schedule of deliverables

All deadlines are 11:59 pm Central time unless a time is given.

- Homework 3 (transport layer): due Friday, November 13, 2026.
- Quiz 4 (TCP congestion control): Wednesday, November 18, 2026, in class at 4:00 pm.
- Networks project Part 3 (congestion control and evaluation): due Friday, November 20, 2026.
- Homework 4 (routing algorithms): due Wednesday, December 2, 2026.
- Homework 5 (link layer): due Wednesday, December 9, 2026.
- Final exam: Thursday, December 10, 2026, 3:30 pm, Royall Hall 111 (cumulative).

Weekly topics

- Week 10 (Nov 2): Module 4 - Transport layer: TCP reliability and flow control
- Week 11 (Nov 9): Module 4 - TCP congestion control (slow start, AIMD, fast recovery)
- Week 12 (Nov 16): Module 5 - Network layer: IP addressing and forwarding
- Week 13 (Nov 23): Thanksgiving break, no class
- Week 14 (Nov 30): Module 5 - Routing algorithms: link state and distance vector
- Week 15 (Dec 7): Module 6 - Link layer and review

Policies

Late work loses 10% per day, up to three days. Collaboration on concepts is encouraged; code and written answers must be your own. AI study tools may be used to understand material but not to produce graded answers. RooRoom's Study AI will give hints, not answers, on graded work.